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PS C:\Users\molef> curl.exe -v https://httpbin.org/get
* Trying 32.196.65.112...
* TCP_NODELAY set
* Connected to httpbin.org (32.196.65.112) port 443 (#0)
* schannel: SSL/TLS connection with httpbin.org port 443 (step 1/3)
* schannel: checking server certificate revocation
* schannel: sending initial handshake data: sending 176 bytes...
* schannel: sent initial handshake data: sent 176 bytes
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: failed to receive handshake, need more data
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: encrypted data got 4096
* schannel: encrypted data buffer: offset 4096 length 4096
* schannel: encrypted data length: 185
* schannel: encrypted data buffer: offset 185 length 4096
* schannel: received incomplete message, need more data
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: encrypted data got 162
* schannel: encrypted data buffer: offset 347 length 4096
* schannel: sending next handshake data: sending 126 bytes...
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: encrypted data got 204
* schannel: encrypted data buffer: offset 204 length 4096
* schannel: SSL/TLS handshake complete
* schannel: SSL/TLS connection with httpbin.org port 443 (step 3/3)
* schannel: stored credential handle in session cache
> GET /get HTTP/1.1
> Host: httpbin.org
> User-Agent: curl/7.55.1
> Accept: */*
>
* schannel: client wants to read 102400 bytes
* schannel: encdata_buffer resized 103424
* schannel: encrypted data buffer: offset 0 length 103424
* schannel: encrypted data got 544
* schannel: encrypted data buffer: offset 544 length 103424
* schannel: decrypted data length: 230
* schannel: decrypted data added: 230
* schannel: decrypted data cached: offset 230 length 102400
* schannel: encrypted data length: 285
* schannel: encrypted data cached: offset 285 length 103424
* schannel: decrypted data length: 256
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Windows PowerShell
> Host: httpbin.org
> User-Agent: curl/7.55.1
> Accept: */*
>
* schannel: client wants to read 102400 bytes
* schannel: encdata_buffer resized 103424
* schannel: encrypted data buffer: offset 0 length 103424
* schannel: encrypted data got 544
* schannel: encrypted data buffer: offset 544 length 103424
* schannel: decrypted data length: 230
* schannel: decrypted data added: 230
* schannel: decrypted data cached: offset 230 length 102400
* schannel: encrypted data length: 285
* schannel: encrypted data cached: offset 285 length 103424
* schannel: decrypted data length: 256
* schannel: decrypted data added: 256
* schannel: decrypted data cached: offset 486 length 102400
* schannel: encrypted data buffer: offset 0 length 103424
* schannel: decrypted data buffer: offset 486 length 102400
* schannel: schannel_recv cleanup
* schannel: decrypted data returned 486
* schannel: decrypted data buffer: offset 0 length 102400
< HTTP/1.1 200 OK
< Date: Sun, 05 Jul 2026 19:02:47 GMT
< Content-Type: application/json
< Content-Length: 256
< Connection: keep-alive
< Server: gunicorn/19.9.0
< Access-Control-Allow-Origin: *
< Access-Control-Allow-Credentials: true
<
{
  "args": {},
  "headers": {
    "Accept": "*/*",
    "Host": "httpbin.org",
    "User-Agent": "curl/7.55.1",
    "X-Amzn-Trace-Id": "Root=1-6a4aaa57-60a4050b7f886f6f4f2e7185"
  },
  "origin": "197.184.93.128",
  "url": "https://httpbin.org/get"
}
* Connection #0 to host httpbin.org left intact
PS C:\Users\molef> curl.exe -v https://httpbin.org/get
```

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Windows PowerShell
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Try the new cross-platform PowerShell https://aka.ms/pscore6

PS C:\Users\molef> curl.exe -v --ssl-no-revoke https://httpbin.org/get 2>&1 | findstr /i "cipher ssl tls suite"
+ curl.exe -v --ssl-no-revoke https://httpbin.org/get 2>&1 | findstr /i ...
* schannel: SSL/TLS connection with httpbin.org port 443 (step 1/3)
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: SSL/TLS connection with httpbin.org port 443 (step 2/3)
* schannel: SSL/TLS handshake complete
* schannel: SSL/TLS connection with httpbin.org port 443 (step 3/3)
PS C:\Users\molef> curl.exe -o NUL -s -w "DNS:   %{time_namelookup}s\nTCP:   %{time_connect}s\nTLS:   %{time_appconnect}s\nFirst byte: %{time_starttransfer}s\nTotal:   %{time_total}s\n" https://httpbin.org/get
DNS:   0,094000s
TCP:   0,360000s
TLS:   0,891000s
First byte: 1,141000s
Total: 1,157000s
PS C:\Users\molef>
```

- The above screenshot displays the machine sending a request to an http server. The above request makes use of TLS communication protocol. The machine uses **schannel** which works on windows devices. On mac and Linux OS devices the TLS version used is OpenSSL. The TLS version is the **TLS 1.2**.

- The Cipher Suite explains the protocol being used to handle the request as well as the encryption, the mode and the hash function used for data integrity. The Cipher Suite being utilised for this request is **TLS/SSL**
- Looking at the cache header and we can see that there are no cache header and this is a deliberate answer. Dynamic API responses are treated as not cacheable by default so expecting a cache header for this request would not make sense.
- The HTTP status code appears after the HTTP/1.1, in our case the HTTP status code used in our request is **200**. The **OK** serves to show that the connection was successful.
- The time it takes to establish a connection can be describes using the
DNS: lookup 94ms Time taken to resolve httpbin.org IP Address
TCP: $0.360 - 0.094 = 266\text{ms}$: Establish 3 way SYN/SYN-ACK/ACK to open the connection
TLS Overhead: $0.891 - 0.360 = 531\text{ms}$: This represent the time taken to exchange the certificate and ensure the cipher negotiation.
First byte: $1.141 - 0.891 = 250\text{ms}$: Time taken for the serve to receive requests and sent the first byte as a response to the request
Total: $1.157 - 1.141 = 16\text{ms}$: Downloading the actual response body

